26900141

MAHATMA GANDHI UNIVERSITY ,KOTTAYAM
MGU-UGP (Honours) /MGU-BCA (Honours)
REGULAR EXAMINATION October 2026
Fifth Semester
Core Course(CC)-MG5CCRBCA300 –COMPUTER NETWORKS
(2024 ADMISSION ONWARDS)
Scheme of Evaluation

Duration 1 hour 30 minutes **Max Marks 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A
Very Short Answer Questions
Answer all questions
Each question carries 2 marks

1. Define a digital signal [K], [CO1,[CO2]

Ans. A digital signal is a type of signal that represents data using a sequence of discrete , separate values rather than a smooth continuous range.

2. List any four networking devices [K]/[CO1]

Ans. Repeaters, Bridges, Routers, Hub, Gateways etc.

3. Differentiate Circuit switching and packet switching [U]/[CO1]

Ans. A circuit-switched network consists of a set of switches connected by physical links. A connection between two stations is a dedicated path made of one or more links. However, each connection uses only one dedicated channel on each link. Each link is normally divided in to **n** channels by using FDM or TDM.

In data communications, we need to send messages from one end system to another. If the message is going to pass through a packet-switched network, it needs to be divided into packets of fixed or variable size. The size of the packet is determined by the network and the governing protocol. In packet switching, there is no resource allocation for a packet. This means that there is no reserved bandwidth on the links, and there is no scheduled processing time for each packet.

4 .Explain the significance of NAK control message [U]/[CO2]

Ans. A NAK control message is a signal sent by a receiver to a sender to indicate that a data packet or transmission was corrupted , missing or unreadable.

5. Explain CSMA [U]/[CO2]

Ans. To minimize the chance of collision and, therefore, increase the performance, the CSMA method was developed. The chance of collision can be reduced if a station senses the medium before trying to use it. Carrier sense multiple access (CSMA) requires that each station first listen to the medium (or check the state of the medium) before sending. In other words, CSMA is based on the principle "sense before transmit" or "listen before talk." CSMA can reduce the possibility of collision, but it cannot eliminate it.

6. Mention any two functions of the Data link layer [U]/[CO2]

Ans. The most important responsibilty of data link layer are flow control and error control. Other responsibilities are :- Framing, Physical addressing, Flow control, Error control, Access control.

7. A computer knows the IP address of the destination but not its physical address .which protocol do address resolution? [A]/[CO2]

Ans. The protocol responsible for finding a devices physical address (MAC address) when its IP address is known the Address resolution Protocol(ARP).IP address responds directly to the source computer with its Physical address.

8. Define network latency [K]/[CO3]

Ans. Network latency is the time delay it takes for a data packet to travel from its source to its destination across a network usually measured in milliseconds.

9. Explain two components of an HTTP request message [U]/[CO3]

Ans. The HTTP is a protocol used mainly to access data on the World Wide Web. HTTP functions as a combination FTP an SMTP. The HTTP messages are read and interpreted by the HTTP server and HTTP client (browser). The HTTP transaction between the client and server. A HTTP uses the services of TCP, HTTP itself is a stateless protocol. The client initializes the transaction by sending a request message. The server replies by sending a response. Messages The formats of the request and response messages are similar. A request message consists of a request line, a header, and sometimes a body. A response message consists of a status line, a header, and sometimes a body.

10. Explain network management [U]/[CO3]

Ans. Network management refers to the administrative , technical, and operational processes used to monitor ,maintain ,secure and optimize a computer network ,its primary goal is to ensure that a network runs efficiently ,reliably and securely while meeting the performance needs of its users.

Part B
Short answer questions
Answer any 3 questions
Each question carries 5 marks

11. Describe with examples how computer networks are used in real time applications like video Conferencing and online gaming [U]/[CO1]

Ans. Real time applications demand immediate data transmission with minimal delay. unlike downloading a file where a few seconds of delay goes unnoticed, applications like video conferencing and online gaming require data to flow seamlessly in fraction of a second.

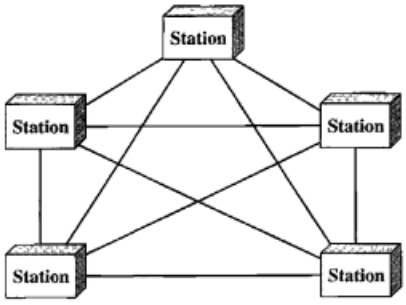
Video conferencing turns analog audio and video into digital streams , compress them and sends them across the globe via computer networks.

Online gaming requires constant , lightening fast communication between a players device and a central game server to synchronize actions.

12. Explain mesh topology with a diagram and discuss its advantages and disadvantages.

[U]/[CO1]

Ans. In a mesh topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects. To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node. Node 1 must be connected to n - I nodes, node 2 must be connected to n - 1 nodes, and finally node n must be connected to n - 1 nodes



A mesh offers several advantages over other network topologies. First, the use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.

The main disadvantages of a mesh are related to the amount of cabling and the number of I/O ports required. First, because every device must be connected to every other device, installation and reconnection are difficult.

13. Explain the IEEE standards used for Wired LANs and describe their major characteristics

Ans. IEEE standards define the physical layer and data link layer specifications for networking technologies.LAN s standards is defined under IEEE 802.3 working group commoly known as Ethernet.

Ethernet Evolution Through four generations- Standard Ethernet(10Mbps),
Fast Ethernet (100 Mbps), Gigabit Ethernet(1 Gbps), Ten gigabit Ethernet(10 Gbps) [U/[CO2]

14. Describe the IPv4 addressing scheme [U]/[CO2]

Ans. A protocol such as IPv4 that defines addresses has an address space, There are two notations to show an IPv4 address-Binary and dotted decimal notation.IPv4 Address is a unique 32 bit numerical identifier assigned to every device connected to a computer network that uses the internet protocol for communication. written in dotted decimal notation consisting of four 8 bit octets separated by dots.(Eg.117.149.29.2). In binary notation the IPv4 address is displayed in 32 bits .(Eg. 01100011 00000010 00011101 10010101)

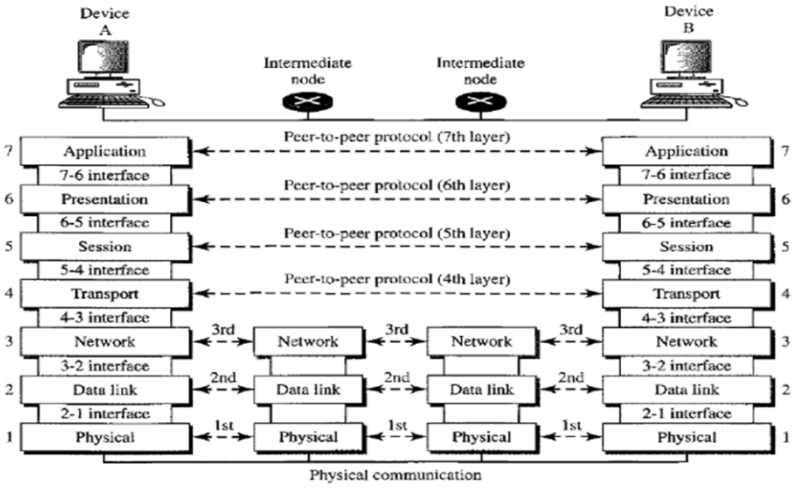
15. Give the primary objectives of network monitoring [K]/[CO3]

Ans. A network must be able to meet a certain number of criteria. The most important of these are performance, reliability, and security. [5 X 3=15]

Part C
Essay Questions
Answer any 1 questions
Each question carries 15 marks

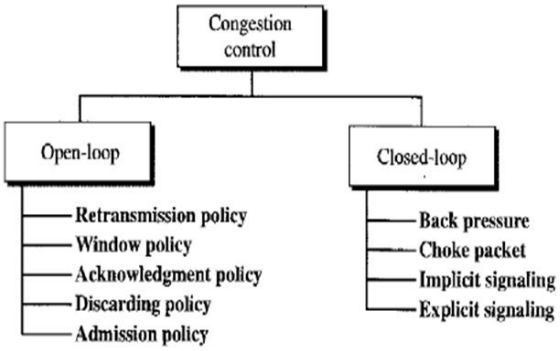
16. With the help of neat diagram explain the functions of each layer in OSI reference Model. [K]/[CO2]

Ans. Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection model. It was first introduced in the late 1970s. An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture. The purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software .



17. Explain the major TCP congestion control techniques in detail .Discuss the working of slow start , Congestion Avoidance ,File retransmit, and Fast Recovery process [U]/[CO3]

Ans. Congestion control refers to techniques and mechanisms that can either prevent congestion, before it happens, or remove congestion, after it has happened. In general, we can divide congestion control mechanisms into two broad categories: open-loop congestion control (prevention) and closed-loop congestion control (removal)



In open-loop congestion control, policies are applied to prevent congestion before it happens. In these mechanisms, congestion control is handled by either the source or the destination.

In Closed-loop congestion control mechanisms try to alleviate congestion after it happens. Several mechanisms have been used by different protocols.

[15 X 1=15]

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